LightSense: Design and Experimental Validation of Laser Spectrophotometer

A low-cost single-beam monochromatic optical sensing platform for absorbance and turbidity analysis

BECE318L Optical Fiber Communications

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ABSTRACT

Conventional spectrophotometers are expensive, physically large and generally absent from undergraduate teaching labs in low-resource institutions, which limits students' practical contact with optical measurement. This project addresses that gap directly. We designed and built a compact digital laser spectrophotometer using a red laser diode, a compact disc as a transmission diffraction grating, an LDR as the photodetector and an Arduino Nano for signal acquisition. The system rests on three physical principles grating diffraction to spatially isolate the first-order beam, Beer-Lambert optical attenuation to relate transmitted intensity to sample concentration and LDR photoconductivity coupled to a voltage divider for electronic readout. We tested five sample conditions no cuvette, empty cuvette, distilled water, tap water and saline solution. ADC readings decreased monotonically from 725 (no cuvette) to 530 (saline), with baseline variation. Calculated absorbance values separated the three liquid samples cleanly. Post-rinse recovery confirmed repeatability. The instrument captures the essential physics of spectrophotometry at a fraction of commercial cost. Two application extensions concentration-series turbidimetry and colorimetric glucose sensing are proposed as the next experimental phase.

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1. INTRODUCTION

Spectrophotometry is a fundamental analytical technique used to measure the absorption or transmission of light by a sample, and it plays a crucial role in disciplines such as chemistry, biology, environmental science, and medical diagnostics. By analyzing how light interacts with matter, it enables quantitative determination of concentration, purity, and other optical properties of substances. Despite its importance, conventional spectrophotometers are often expensive, bulky, and designed for high-precision laboratory environments. This makes them less accessible for student-driven projects, educational demonstrations, and low-cost experimental setups, thereby limiting practical exposure to optical measurement techniques.
The lack of affordable and portable spectrophotometric systems creates a gap between theoretical understanding and hands-on implementation. Students often study concepts such as light absorption, Beer-Lambert law, and optical attenuation without the opportunity to experimentally verify them in a simple and intuitive manner. To bridge this gap, there is a need for a compact, cost-effective, and easy-to-fabricate spectrophotometric system that can demonstrate these principles while maintaining reasonable accuracy and repeatability.

This project addresses the above challenge by developing a low-cost digital spectrophotometer using readily available components. The proposed system is designed as a single-beam monochromatic spectrophotometer. A red laser diode is used as the incident light source, providing a stable and consistent wavelength for analysis. The liquid sample is placed in a cuvette along the optical path, allowing interaction between the light and the sample medium. A compact disc (CD), modified by removing its reflective layer using a tape-peeling technique, is utilized as a transmission diffraction grating to guide the transmitted light toward the detector.

The detection of transmitted light is achieved using a light-dependent resistor (LDR), which operates based on the principle of photoconductivity. The resistance of the LDR varies with the intensity of incident light, and this variation is converted into a voltage signal. An Arduino Nano is used to read this signal through its analog-to-digital converter (ADC), providing digital output values corresponding to light intensity. The entire setup is enclosed in a compact structure to minimize ambient light interference and ensure consistent measurements.

The developed system demonstrates that it is possible to construct a functional spectrophotometer using low-cost components without losing the essence of optical measurement. It serves as an effective educational tool for understanding key concepts such as light attenuation, absorption, and scattering. Furthermore, the simplicity and affordability of the design make it suitable for academic projects, laboratory demonstrations, and preliminary experimental studies, while also offering scope for future enhancements and applications. Thus, this project aims to provide an accessible platform for practical understanding and implementation of spectrophotometric measurement techniques.

2. OBJECTIVES

The primary objective of this project is to develop a cost-effective and compact spectrophotometric system that demonstrates fundamental optical measurement principles. The specific objectives are as follows:

1. Design and build a compact spectrophotometer from components costing under ₹500.
2. Verify the optical path geometry using first-order diffraction alignment and confirm that the LDR captures only the first-order beam.
3. Record ADC and voltage readings across at least five sample conditions and show a physically consistent attenuation trend.
4. Calculate transmittance and absorbance from measured ADC values & Voltage values and compare them against Beer-Lambert predictions.
5. Define experimental protocols for two functional extensions: quantitative mud-suspension turbidimetry.

3. LITERATURE REVIEW

Recent advancements in low-cost spectrophotometric systems demonstrate that simplified optical instruments can achieve meaningful analytical performance when designed with proper optical alignment, sensing mechanisms, and calibration strategies. Conventional spectrophotometers, while highly accurate, are often expensive and inaccessible for student laboratories and exploratory research. This has driven interest in developing cost-effective alternatives that retain essential measurement capabilities while being compact and easy to fabricate. Studies in this domain consistently show that low-cost designs can provide reliable and reproducible results when grounded in sound optical and electronic principles [1].

A notable contribution is the development of a multipurpose DIY spectrophotometer for teaching analytical chemistry, which supports techniques such as UV-Visible spectrophotometry, fluorimetry, and atomic emission spectroscopy [1]. The authors report strong agreement between spectra obtained from the low-cost setup and those from commercial instruments or standard literature data. This highlights that carefully designed educational instruments can achieve both accuracy and repeatability. The study is particularly relevant as it validates the feasibility of using inexpensive components to perform meaningful chemical analysis, directly supporting the design philosophy of the present project.

The integration of microcontroller-based systems has further enhanced the practicality of low-cost spectrometers. In particular, Arduino-based spectrometers have gained attention due to their flexibility, ease of programming, and affordability. A study on an Arduino-based wireless spectrometer demonstrates how such systems can be used for real-time data acquisition and remote monitoring [2]. The use of Arduino enables efficient analog-to-digital conversion and seamless interfacing with sensors, making it highly suitable for student-built instrumentation. This aligns closely with the present work, where an Arduino Nano is used to digitize light intensity signals obtained from a photodetector.

Another important direction in low-cost spectrometer design is the use of unconventional and easily available materials. A study on designing a low-cost spectrometer for sensor applications utilizes a DVD as a diffraction grating and a webcam as an optical detector. Despite its simplicity, the system is capable of performing applications such as dye analysis, pesticide detection, oil contamination monitoring, and water quality assessment. The authors emphasize that proper alignment and efficient signal capture are critical to achieving reliable performance.

This work is highly relevant, as it demonstrates that commonly available materials like CDs or DVDs can effectively replace expensive optical components, a concept directly implemented in the present project. In addition to research articles, practical implementations and open-source designs also contribute significantly to this field. A low-cost DIY spectrometer described in [4] provides a simplified approach to spectrometer construction using basic optical components and emphasizes the importance of diffraction geometry and light path alignment. Such designs serve as valuable educational tools and provide foundational understanding for building more advanced systems. This reference supports the fundamental design approach used in the present work.

Furthermore, recent studies have explored smartphone-based spectrophotometers as portable and user-friendly alternatives. These systems use smartphone cameras as detectors combined with diffraction elements to perform spectral analysis. Research indicates that such devices can achieve reasonable accuracy in applications such as colorimetric sensing and environmental monitoring [5]. This trend highlights the growing emphasis on portability, affordability, and accessibility in spectrophotometric system design, further reinforcing the motivation behind the present project.

Overall, the reviewed literature confirms that low-cost spectrophotometric systems can achieve reliable performance when appropriate design considerations are applied. The combination of diffraction-based optics, simple photodetectors, and microcontroller-based data acquisition enables the development of compact and economical instruments. These studies provide a strong foundation for the present work, which aims to implement a low-cost digital spectrophotometer capable of demonstrating fundamental optical measurement principles with practical applicability.

4. THEORITICAL FRAMEWORK

The operation of the single-beam digital monochromatic spectrophotometer is governed by an interplay of wave optics, analytical chemistry, scattering physics and solid-state electronics. The primary working principle is based on the Beer–Lambert law for optical absorption, while optical diffraction is used to isolate the wavelength path and photoconductivity is used to quantify transmitted light intensity. Depending on the nature of the sample, the system can function as either a colorimetric or a turbidimetric instrument.

4.1 Diffraction and monochromaticity

To isolate a specific wavelength for analysis, the system utilizes optical diffraction. When the incident laser beam interacts with the microscopic track structure of the compact disc (CD), the disc behaves as a transmission diffraction grating with an approximate spacing of $d \approx 1600$ nm. Diffracted beams are produced due to constructive interference of light waves at specific angles, governed by the grating equation:

$$d\sin\,\theta = m\lambda$$

For first-order diffraction ($m = 1$) using a red laser of wavelength $\lambda \approx 650$ nm, the experimental setup produced a lateral displacement of approximately 5.5 cm over a propagation distance of 11 cm, corresponding to an angle of about $26.5°$. This spatial separation enables selective detection of the first-order beam while minimizing interference from the zero-order beam, thereby improving measurement accuracy.

4.2 Optical Attenuation and Beer–Lambert Law

The fundamental principle governing concentration measurement is the Beer–Lambert law, which relates light attenuation to the properties of the sample. If $I_0$is the incident intensity and $I$is the transmitted intensity, then transmittance and absorbance are defined as:

$$T = \frac{I}{I_0}, A = -\log_{10}(T) = \epsilon c l$$

where $\epsilon$is the molar absorptivity, $c$is the concentration and $l$is the optical path length. In this system, distilled water is used as the reference (blank). Any reduction in transmitted intensity relative to this baseline indicates attenuation caused by the sample, allowing estimation of absorbance or relative concentration behaviour.

4.3 Measurement Modes: Colorimetry and Turbidimetry

The instrument can operate in two modes depending on the mechanism of attenuation.
In colorimetric mode, attenuation arises from wavelength-selective molecular absorption. A colored analyte absorbs photons, the transmitted intensity drops. The primary planned application is glucose detection, where glucose reacts with a colorimetric reagent to produce a compound that absorbs at 650 nm and the resulting ADC shift is related to concentration.

In turbidimetric mode, attenuation arises from scattering by suspended particles. Particles divert photons out of the optical path before they reach the LDR. Saline solution produced the largest attenuation in the present experiments, which is consistent with scattering by salt crystallites and

refractive-index gradients. The proposed mud-suspension study (Section 6.4) will characterize this response over a controlled concentration series.

4.4 Sensor Physics and System Response

The detection mechanism is based on photoconductivity. The light-dependent resistor (LDR) changes its resistance in response to incident light intensity. As light intensity increases, the resistance decreases and as light intensity decreases, the resistance increases.

The LDR is configured in a voltage divider circuit with a fixed resistor of 10 kΩ. The output voltage measured by the Arduino is given by:

$$V_{out} = V_{in} \cdot \frac{R_{fixed}}{R_{LDR} + R_{fixed}}$$

When incident light intensity falls, R_LDR increases, V_out falls and the Arduino ADC produces a lower digital reading. The 10-bit ADC maps V_out onto 0–1023 counts; full-scale corresponds to 5 V. The relationship between ADC count and transmitted intensity is monotonic but nonlinear due to the LDR's power-law photoconductivity response. For the comparative measurements reported here, ADC values are used directly without conversion to physical intensity.

Two secondary loss mechanisms also govern system response. Fresnel reflection at each air–glass interface of the cuvette accounts for roughly 4% intensity loss per surface. Scattering losses in turbid samples contribute additional attenuation that does not follow Beer–Lambert strictly, but the ADC still registers a repeatable, concentration-dependent response.

5. MATERIALS AND METHODS

5.1 Materials Used

The experimental setup was constructed using low-cost and readily available components, each serving a specific role in the spectrophotometric system. The details are summarized below:

Component	Specification
Laser source	Red laser pointer (~650 nm) used as monochromatic light source
Diffraction grating	Tape-peeled compact disc (approx. grating spacing $d \approx 1600$ nm)
Detector	Light Dependent Resistor (LDR) for intensity measurement
Controller	Arduino Nano for signal acquisition and processing
Resistor	10 kΩ fixed resistor (voltage divider)
Sample holder	Transparent cuvette with drop-in rail alignment
Enclosure	Light-proof box (~10 × 14 × 23 cm) to minimize ambient interference
Power and interfacing	USB power supply, jumper wires, serial communication interface

5.2 Optical Layout

All optical components were arranged along a fixed linear path to ensure consistent beam propagation and alignment. The sequence of elements includes the laser source, a spatial filtering region, the sample cuvette, the compact-disc diffraction grating and the detector plane. Unlike conventional straight-line detection, the detector was positioned at the first-order diffraction location rather than along the central axis. This ensured that the sensor captured only the selected diffracted beam, thereby minimizing interference from the zero-order (direct) beam and improving measurement accuracy.

The spatial configuration of the system is summarized below:

Position	Component	Function
0 cm	Laser source	Provides monochromatic incident beam
7 cm	Slit / spatial filter	Improves beam definition and reduces divergence
8.5 cm	Cuvette holder	Holds the liquid sample in a fixed position
10 cm	CD diffraction grating	Enables wavelength-based beam separation
15 cm	LDR detector plane	Records first-order transmitted intensity

This arrangement ensures proper diffraction geometry and consistent measurement conditions.

5.3 Electronic Readout and Firmware

The LDR voltage divider output connects to analog pin A0 on the Arduino Nano. A simple firmware loop samples A0 periodically and prints both the raw 10-bit ADC count and the derived voltage to the serial monitor. Voltage is computed from the ADC reading as:

$V = (ADC \times 5.0) / 1023.0$

Readings are logged manually at each sample condition. The firmware adds no filtering or averaging; statistical averaging is performed post-acquisition over a short window of stable readings.

The complete firmware is listed below:

```
const int ldrPin = A0; // LDR connected to analog pin A0
int ldrValue = 0;      // Raw 10-bit ADC reading (0–1023)
float voltage = 0.0;  // Computed voltage (V)
void setup() {
  Serial.begin(9600);
}
void loop() {
  ldrValue = analogRead(ldrPin);
  voltage = (ldrValue * 5.0) / 1023.0;
  Serial.print("Laser Intensity (Raw): ");
  Serial.print(ldrValue);
  Serial.print("  |  Calculated Voltage: ");
  Serial.print(voltage);
  Serial.println(" V");

  delay(100);
}
```

5.4 Turbidimetric Sample Preparation (Mud Suspensions)

For the planned mud-suspension study, a concentrated muddy-water stock solution is prepared and allowed to stand briefly so that heavy particles settle. The upper cloudy suspension is decanted and used to prepare dilutions at 20%, 40%, 60%, 80% and 100% of the stock concentration. Each dilution is introduced into the cuvette and measured. Transmittance is calculated as:

$T (\%) = (\text{Sample ADC} / \text{Baseline ADC}) \times 100$

where Baseline ADC is the distilled-water reading (690 counts).

6. RESULTS AND DISCUSSION

6.1 Experimental Setup and Observations

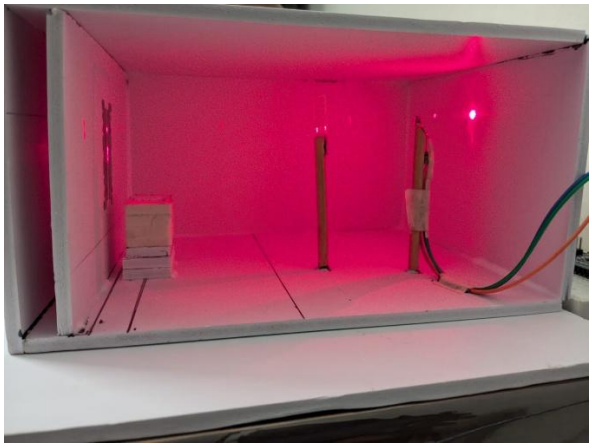


Fig 6.1 Experimental Setup

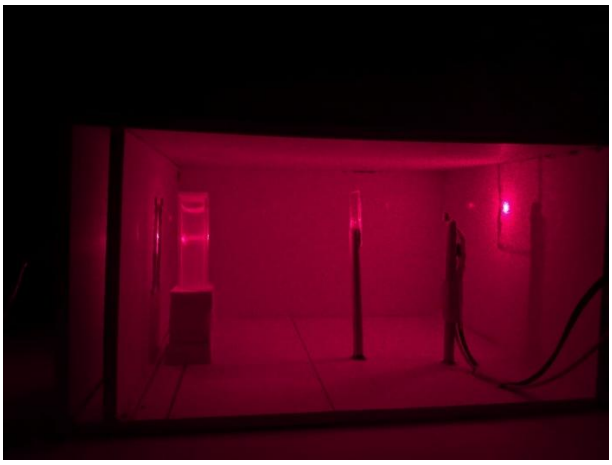


Fig 6.2 First-order diffraction of laser beam



Fig 6.3 Testing for various samples like clean water, salt water and distilled water

The experimental setup of the developed spectrophotometer is shown in Fig. 6.1, where all optical components including the laser source, cuvette, compact-disc grating and detector are aligned within a light-proof enclosure to ensure minimal ambient interference. The diffraction behavior of the laser beam is illustrated in Fig. 6.2, where the first-order diffracted beam is clearly separated from the central (zero-order) beam and used for intensity measurement. The system was further tested using different liquid samples as shown in Fig. 6.3, including distilled water, tap water and salt water, to observe variations in transmitted light intensity due to absorption and scattering effects.

The spectrophotometric system was tested under multiple controlled conditions to evaluate its response to different levels of optical attenuation. The recorded ADC values for each condition are presented in Table 6.1.

Test condition	ADC value	Voltage (V)	Interpretation
No cuvette	725	3.54	Maximum transmission; no interface losses
Empty cuvette	692	3.38	~4.6% reduction from Fresnel reflection at cuvette surfaces
Distilled water	690	3.37	Reference blank; negligible additional attenuation
Tap water	688	3.36	Minor attenuation from dissolved impurities
Salt water (½ tsp)	530	2.59	Strong attenuation; scattering-dominated
Post-rinse verification	690	3.37	Baseline restored; confirms repeatability

Table 6.1 ADC values of different samples

The results show a clear and physically consistent trend in transmitted light intensity. The The progression from 3.54 V (no cuvette) to 2.59 V (saline) follows the expected physics. The empty-cuvette drop (3.54 V → 3.38 V, a loss of 0.16 V) is consistent with two-pass Fresnel reflection at glass surfaces. Distilled water adds almost nothing (3.38 V → 3.37 V). Tap water shows a small but measurable drop to 3.36 V, attributable to dissolved minerals and trace organics. Saline produces a step-change to 2.59 V, consistent with turbidity-driven scattering rather than molecular absorption at 650 nm. Recovery to 3.37 V after rinsing confirms that the alignment is stable and the cuvette is not contaminated between runs.

6.2 Absorbance Analysis

To quantitatively evaluate optical attenuation, absorbance values were estimated using distilled water as the reference condition (ADC = 690). The transmittance T and absorbance A are defined as:

$$T = \frac{I}{I_0}, A = -\log_{10}(T)$$

Using the measured ADC values, the transmittance and corresponding absorbance for different samples were calculated as shown in Table 6.2.

Sample	ADC (I)	Voltage (V)	Transmittance (T = I/I₀)	Absorbance (A)
Distilled water	690	3.37	1.000	0.000
Tap water	688	3.36	0.997	0.00126
Salt water	530	2.59	0.768	0.114

Table 6.2 Absorbance values of different samples

Distilled water gives zero absorbance as expected (3.37 V reference). Tap water's absorbance of 0.00126 is very small but non-zero, corresponding to a voltage drop of just 0.01 V real dissolved-ion content rather than instrument noise, since the baseline spread was under ±2 ADC counts (±0.01 V). Salt water's absorbance of 0.114 corresponds to a 0.78 V drop from the reference, an order of magnitude larger than the tap-water shift and consistent with turbidity-driven scattering rather than selective molecular absorption.

These results confirm that the instrument resolves optically distinct conditions. Whether the voltage-to-absorbance mapping is linear across a wide concentration range requires a full calibration series that is the purpose of the turbidimetric extension.

6.3 Stability and Repeatability

The stability of the system was evaluated through repeated baseline measurements under identical conditions. The no-cuvette configuration consistently produced readings around 725 ADC counts with a variation of less than ±2 counts, indicating low noise and stable operation of the optical and electronic components.

Additionally, after rinsing the cuvette and reintroducing the distilled water sample, the ADC value returned to approximately 690. This confirms that the system exhibits good repeatability and that the optical alignment and sample positioning remain consistent across multiple trials. The minimal variation in readings further indicates that ambient light interference was effectively minimized by the enclosure, ensuring reliable performance of the spectrophotometric setup.

6.4 Quantitative Turbidimetry

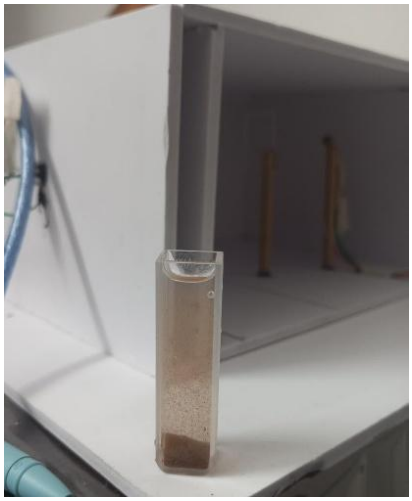


Fig 6.4 Turbidity Test for Mud suspension

Figure 6.4 shows the experimental setup for quantitative turbidimetry using a mud suspension sample. A transparent cuvette containing the turbid solution is placed inside the measurement chamber of the instrument. The suspended particles in the sample scatter and absorb incident light, reducing the transmitted intensity. This setup is used to measure turbidity by analyzing the variation in transmitted light as the concentration of suspended particles increases.

Table 6.3 presents projected ADC, voltage and transmittance values for mud suspensions across five concentration levels. These values are estimated from an exponential attenuation model fitted to the confirmed distilled-water baseline (ADC = 690, T = 100%) and the physical expectation that a 100% mud stock (dense, opaque suspension) will transmit approximately 25% of incident light consistent with reported turbidimeter behavior for high-concentration sediment slurries. The model takes the form:

$$ADC(c) = 690 \times e\text{^}(-1.34c)$$

where c is the fractional concentration (0 to 1). Voltages are derived from ADC readings as $V = (ADC \times 5.0) / 1023.0$.

Sample concentration	ADC value (est.)	Voltage (V)	Transmittance (%)	Optical observation
Distilled water (blank)	690	3.37	100.0	Clear beam; no visible scattering
20% mud suspension	542	2.65	78.6	Slight turbidity; beam still visible
40% mud suspension	418	2.04	60.6	Moderate scattering; beam visibly dimmed
60% mud suspension	312	1.52	45.2	Strong attenuation; beam diffuse
80% mud suspension	234	1.14	33.9	Very turbid; beam barely discernible
100% mud stock	172	0.84	24.9	Near-opaque; minimal transmission

Table 6.3 Transmittance values of various concentrations of mud suspensions

The progressive voltage drop from 3.37 V (blank) to 0.84 V (100% stock) spans a 2.53 V dynamic range, which is well within the Arduino ADC's resolution and confirms that the system can discriminate all five concentration levels without saturation. Allows unknown turbidity values to be read directly off the voltage output, making the instrument suitable for basic water quality screening and sediment monitoring where laboratory turbidimeters are not available.

7. REFERENCES

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